

TON-IoT Processed Datasets Analysis Report

Contextual Intelligence Engine (CIE) Evaluation & Selection

EXECUTIVE SUMMARY

The TON-IoT dataset contains 38 processed telemetry files (~4.8 GB) spanning 4 heterogeneous environments:

1. Processed IoT Telemetry (7 CSV files): Physical sensor and actuator operational data.
2. Processed Linux OS Telemetry (6 CSV files): Host kernel process, disk I/O, and memory usage.
3. Processed Windows OS Telemetry (2 CSV files): Windows 7 & 10 performance counter logs.
4. Processed Network Telemetry (23 CSV files): Bro/Zeek flow connection logs (46 features).

CIE RECOMMENDATION & SELECTION

RECOMMENDED DATASET: Processed Network Telemetry (Network_dataset_*.csv)

Justification for Selection:

- Unified Multi-Dimensional Context: Incorporates 5 context categories (Network, Temporal, Behavioral, and Temporal) into a single unified telemetry schema.
- Testbed-Wide Visibility: Captures inter-device network communications across ALL devices, including Linux servers, Windows workstations, and attack orchestrators.
- Rich Feature Set (46 Features): Combines transport metrics (TCP/UDP), application layer data (HTTP, DNS, SSL), and specialized Zeek security anomaly flags ('weird_name', 'weird_status', etc.).
- Precise Ground-Truth Targets: Features both binary classification ('label') and multi-class category annotations ('type').

Summary Table of Processed Datasets (38 Files)

Category	File Count	Total Size	Feature Count	Target Columns	CIE Role
Processed IoT Data	7 CSVs	174.4 MB	6 - 8 cols	label, type, state	Device Context
Processed Linux Data	6 CSVs	296.3 MB	9 - 17 cols	label, type, attack	Host Context
Processed Windows Data	2 CSVs	59.5 MB	127 - 135 cols	label, type	Host Context
Processed Network Data	23 CSVs	4,260.6 MB	46 cols	label, type	PRIMARY CIE DATAS

Domain Analysis: Processed IoT Telemetry (7 Devices)

1. IoT Fridge (IoT_Fridge.csv | 23.8 MB | 6 cols):
 - Features: date, time, fridge_temperature (float64), temp_condition (str), label (int64), type
 - Purpose & Context: Monitors smart fridge temperature dynamics; contributes physical Device Context
2. IoT GPS Tracker (IoT_GPS_Tracker.csv | 30.0 MB | 6 cols):
 - Features: date, time, latitude (float64), longitude (float64), label (int64), type (str).
 - Purpose & Context: Tracks vehicle location coordinates; contributes Geolocation Device Context
3. IoT Garage Door (IoT_Garage_Door.csv | 24.6 MB | 6 cols):
 - Features: date, time, sstate (int64), door_state (str), label (int64), type (str).
 - Purpose & Context: Monitors actuator open/close state; contributes Actuator Behavioral Context
4. IoT Modbus ICS (IoT_Modbus.csv | 14.6 MB | 8 cols):
 - Features: date, time, FC1_Read_Input_Register, FC2_Read_Discrete_Inputs, FC3_Read_Holding_Regi
FC4_Read_Coil, label (int64), type (str).
 - Purpose & Context: Monitors Modbus SCADA industrial control registers; contributes Industrial
5. IoT Motion Light (IoT_Motion_Light.csv | 15.8 MB | 6 cols):
 - Features: date, time, motion_status (int64), light_status (str), label (int64), type (str).
 - Purpose & Context: Monitors motion sensor and light status; contributes Environmental Context.
6. IoT Thermostat (IoT_Thermostat.csv | 18.2 MB | 6 cols):
 - Features: date, time, current_temperature (float64), thermostat_status (int64), label, type.
 - Purpose & Context: Climate control temperature telemetry; contributes Physical Environmental C
7. IoT Weather Station (IoT_Weather.csv | 40.1 MB | 7 cols):
 - Features: date, time, temperature, humidity, pressure, label, type.
 - Purpose & Context: Environmental station sensor readings; contributes Multi-Sensor Environment

Domain Analysis: Linux & Windows OS Telemetry

PROCESSED LINUX TELEMETRY (6 Files | 296.3 MB)

- • Linux Process (Linux_process_1.csv & 2.csv | 17 cols):
Features: ts, PID, TRUN, TSL, TIDL, TSN, ZOMB, RUID, EUID, STUID, SECUID, GIUD, EGID, STGID, label, type.
Context: Process execution states (running, sleeping, zombie) and real/effective user IDs (privileges).
- Linux Disk (linux_disk_1.csv & 2.csv | 9 cols):
Features: PID, RDDSK, WRDSK, WCANCL, DSK, CMD, attack, type.
Context: Disk block read/write volume and executing command strings.
- Linux Memory (linux_memory1.csv & 2.csv | 13 cols):
Features: PID, MINFLT, MAJFLT, VSTEXT, VSIZE, RSIZE, VGROW, RGROW, MEM, CMD, label, type.
Context: Page faults (minor/major), virtual memory growth, and process RAM allocation.

PROCESSED WINDOWS TELEMETRY (2 Files | 59.5 MB)

- • Windows 10 (windows10_dataset.csv | 127 cols) & Windows 7 (windows7_dataset.csv | 135 cols):
Features: Comprehensive Windows Performance Monitor (PerfMon) metrics across Processor, Process, Memory, LogicalDisk, and Network Interface adapters.
Context: Host resource utilization, DPC rates, handle counts, and committed page table bytes.

Primary Recommendation: Processed Network Dataset

PROCESSED NETWORK TELEMETRY (23 Files | 4.26 GB | 46 Features)

Files: Network_dataset_1.csv through Network_dataset_23.csv

FEATURE BREAKDOWN BY CONTEXT CATEGORY:

1. Network Context: src_ip, src_port, dst_ip, dst_port, conn_state, service, proto.
-> Maps inter-device communications, transport protocols, and socket connection states.
2. Traffic Context: duration, src_bytes, dst_bytes, missed_bytes, src_pkts, src_ip_bytes, dst_pkts, dst_ip_bytes.
-> Measures connection throughput, packet rates, byte ratios, and transfer volumes.
3. Application & Behavioral Context:
 - DNS: dns_query, dns_qclass, dns_qtype, dns_rcode, dns_AA, dns_RD, dns_RA, dns_rejected.
 - HTTP: http_trans_depth, http_method, http_uri, http_version, http_request_body_len, http_response_body_len, http_status_code, http_user_agent, http_orig_mime_types, http_referer.
 - SSL: ssl_version, ssl_cipher, ssl_resumed, ssl_established, ssl_subject, ssl_issuer.-> Captures application-layer payload semantics and user-agent interaction behaviors.
4. Security Context: weird_name, weird_addl, weird_notice.
-> Flags protocol violations, invalid header formats, and security anomalies.
5. Temporal Context: ts (Unix timestamp).
-> Tracks event arrival times, inter-arrival gaps, and temporal burst patterns.
6. Target Labels: label (Binary: 0=Normal, 1=Attack), type (Multi-Class: 10 Attack Types).

CONCLUSION:

The Processed Network Dataset provides the most unified, high-dimensional contextual foundation for training and deploying the AI-based Contextual Intelligence Engine (CIE).